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DETAIL SPECIFICATION

ROPE, FIBROUS, PLAITED, CONTINUOUS POLYESTER FILAMENT WITH A STAPLE WRAP

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fibrous plaited rope having a continuous polyester filament with a staple wrap. The rope is furnished in one type and in the circumference specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of this document are available online at http://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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CORDAGE INSTITUE

CI-1500A	-	Test Methods for Fiber Rope Physical Properties
CI-1500B	-	Test Methods for Fiber Rope Performance Properties

(Copies of these documents are available online at www.ropecord.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.

3.2 Materials. Material used in the rope shall be virgin, continuous filament, heat- and light-resistant polyester filament with a staple wrap (otherwise known as multiplex).

3.3 Construction.

3.3.1 Strand configuration. The rope shall be made of eight strands arranged in four pairs, where one individual strand shall be laid adjacent to the second strand in each pair and shall conform to the requirements specified herein. Individual strands shall be made of one size of single yarn and shall have equal numbers of yarns. The yarns shall be made from group filaments.

3.3.2 Preparation. In the preparation of the constituent rope strands, the pattern “SZ” shall be employed in the twisting of the yarns, and the individual strands of the two pairs, while the components of the remaining pairs shall be twisted in the “ZS” pattern.

3.3.3 Finished rope. The finished rope shall be so constructed that in the interweaving procedure, pairs of strands of the former structure shall be twisted in the “Z” direction, while alternating pairs of the latter structure shall be twisted simultaneously in the “S” direction. Heat setting of the rope or any of its components shall not be permitted. The direction of twist for the yarns, the strands, and the pairs of strands into the rope shall be determined in accordance with 4.2.3.

3.4 Physical requirements. The finished rope shall conform to the physical properties specified in table I when tested as specified in 4.2.3.

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TABLE I. Physical Properties.

Circumference at load "P" nominal (inches)	Tolerance plus or minus (inches)	Load "P" (pounds)	Nominal diameter (inches)	Maximum linear density (pounds per 100 feet at load "P")	Breaking test, minimum (pounds)
¾	1/16	12	¼	2.10	2,080
1	1/16	20	5/16	3.15	2,980
1⅛	1/16	30	⅜	4.35	3,970
1¼	1/16	40	7/16	5.65	5,050
1½	⅛	50	½	7.35	6,400
1¾	⅛	70	9/16	9.35	8,100
2	⅛	90	⅝	12.2	9,900
2¼	⅛	110	¾	15.2	12,200
2½	⅛	140	3/16	18.4	14,500
2¾	⅛	170	⅞	21.5	16,700
3	3/16	200	1	25.2	19,000
3¼	3/16	240	1 1/16	29.6	22,000
3½	3/16	270	1⅛	33.6	25,000
3¾	3/16	310	1¼	37.8	27,500
4	3/16	350	1 5/16	42.5	30,700
4½	¼	450	1½	63.0	37,000

3.4.1 Elongation. The elongation of the ropes shall not exceed 25 percent at 75 percent of the minimum breaking test in table I.

3.5 Finish. No extraneous material shall be added for the purpose of weighting the rope.

3.6 Identification marker. The manufacturer shall identify his product by inserting an identification marker between the braids. The marker shall be completely enveloped by the cover yarns in the strand in which it is enclosed. Unless otherwise specified (see 6.2), the manufacturer's name, year of manufacture, and type of fiber (polyester) shall be clearly printed on the marker. Italic or script type shall not be used. The printing shall not be affected upon exposure to water or mineral oil. This marker shall extend the entire length of the rope.

3.7 Identification tag. In addition to marker requirements specified (see 3.6), each package unit shall have a tag attached to it for identification purposes. The tag shall contain the following information:

a. Stock number.
b. Nomenclature.
c. Specification number.
d. Length.
e. Contract number and date.
f. Date of manufacture (month and year).
g. Contractor's name.

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3.8 Put-up. Unless otherwise specified (see 6.2), the rope shall be furnished without knots or splices on non-returnable reels (spools) not larger than 6 feet in diameter in the lengths specified in table II. The ends of all rope shall be cut off squarely and be securely whipped, taped, or heat-sealed to prevent fraying and untwisting. The reels (spools) shall be wound so that each turn and layer is free from entanglement.

3.9 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

TABLE II. Put-up.

Circumference (inches)	Minimum length (feet)	Approx. wt. 1/ (pounds)
¾	2250	50
1	2250	75
1⅛	1620	75
1¼	1200	72
1½	1200	93
1¾	1200	121
2	1200	155
2¼	1200	193
2½	1200	234
2¾	1200	272
3	1200	320
3½	1200	427
3¾	600	240
4	600	270
4½	600	401

NOTE:
1/ Not a requirement; for information only.

4. VERIFICATION

4.1 First article inspection. When specified (see 6.2), the first article sample submitted in accordance with 3.1 shall be visually inspected in accordance with table III. The sample shall then be tested for physical and chemical requirements in accordance with 4.2.3.

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TABLE III. Visual examination of defects.

Examine	Defect	Classification	
		Major	Minor
Appearance and workmanship	Cut, chafed, or damaged, affecting serviceability	X	
	Kinks, darting yarns [1/], broken, or loose ends, bulged strands, strand knots	X	
	Other than eight strands (four pairs)	X	
	Ends not securely whipped, taped or heat sealed to prevent fraying or untwisting		X
Identification marker (ropes 2″ in circumference and larger)	Omitted, incorrect, illegible	X	
	Italic or script type used	X	
	Not completely covered by cover yarns	X	
	Not as specified	X	
Identification tag	Omitted, incorrect, illegible, insecurely attached	X	
	Not as specified.		X
Cleanliness	Spot or stain, clearly visible [2/]		X
	Objectionable odor		X

NOTES:
[1/] Darting yarns are internal yarns that project through the cover yarns of the strand at intervals along the rope.
[2/] At normal inspection distance (approximately 3 feet).

4.2 Inspection. Unless otherwise indicated, sampling and inspection shall be performed in accordance with 4.2.3.

4.2.1 Material and component inspection. Testing shall be conducted on components in accordance with all requirements of this specification (see table IV).

4.2.2 Examination of the end item for visual defects. The defects listed in table III shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. The unit of product for this examination shall be the reel or spool, as applicable. Ten percent of the gross length contained on each unit or product, but not less than 100 feet, shall be subjected to the visual examination. Examination of the end item for visual defects shall be in accordance with MIL-STD-1916.

4.2.2.1 Defects with regard to length. Defects with regard to length shall be considered to exist if any of the following are determined during inspection:

a. Length of unit greater or less than length specified (see 3.8).

b. Length of unit less than that marked on ticket.

c. Any piece less than 600 feet in length for sizes up to and including ropes of 4½-inch circumference.

4.2.2.2 Defects with regard to winding. Defects with regard to winding shall be considered to exist if any of the following are determined during inspection:

a. Improperly or not firmly wound resulting in kinks, knots, entangling, or slippage during unwinding or otherwise affecting free unhampered unwinding of rope.

b. Knot, splice, or otherwise joining of ends to make a continuous length.

4.2.3 Test methods. The methods of testing shall be as specified in table IV.

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TABLE IV. Test methods.

Characteristics	Requirements	Test method	Number of determinations per individual sample unit 1/	Results reported as
Direction of rope twist	3.3	Visual	1	Pass or fail
Direction of twist (plied yarn)	3.3	Visual		Pass or fail
Direction of twist (strand)	3.3	Visual		Pass or fail
Circumference	3.4	4.2.3.1	3	Pass or fail
Linear density	3.4	4.2.3.2	1	Pass or fail
Breaking strength, ropes	3.4	4.2.3.3	3	Pass or fail
Elongation	3.4.1	4.2.3.4	1	Pass or fail

NOTE:
1/ A random sample of reels shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and acceptance quality limit (AQL), as specified (see 6.2).

4.2.3.1 Circumference. The circumference measurement shall be made in accordance with CI-1500A.

4.2.3.2 Linear density. Linear density (pounds per 100 feet) shall be determined in accordance with CI-1500A.

4.2.3.3 Breaking strength. Breaking strength shall be determined in accordance with CI-1500B except as modified herein. Lengths for breaking strength shall be taken from each sample and spliced at each end with a minimum of five tucks. The inside length of each eye, measured with the sides of the eye in contact, shall be not less than 12 inches. The length between the inner ends of the splices shall be a minimum of 3 feet. Clamps or capstan arrangements may be used in lieu of splices; however, in case of dispute, spliced specimens shall be used.

4.2.3.4 Elongation. Elongation shall be determined in accordance with CI-1500B.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The rope covered by this specification is intended for use as messengers and in-haul and out-haul lines for transfer-at-sea between naval ships of personnel, light freight, and other applications where relative freedom from torque is required in addition to high strength, high elongation, and maximum gripping surface.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Circumference (size) required (see 1.1 and table I).

c. When first article sample is required (see 3.1 and 4.1).

d. When information on identification marker is other than specified (see 3.6).

e. Put-up and length, when other than specified (see 3.8).

f. AQL of 1.0 (see table IV).

g. Selection of applicable levels of packaging and packing (see 5.1).

h. That this rope will be purchased on a price-per-pound basis (net weight).

i. That purchaser will accept, at original weight, any unit that has been shortened or cut for test specimens, if in complete compliance with this specification.

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Use of existing stocks. Current stocks of materials may be used until depleted; future requisitions for material should refer to the current version of this specification.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.6 Subject term (key word) listing.

Breaking strength

Elongation

Splice

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – CQ

Preparing activity:
Navy – SH
(Project 4020-2017-005)

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